

FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 4

AC Network Analysis 1

Principles and Applications of Electrical Engineering, 7th ed., Chapter 3
Electric Circuits, 11th ed., Chapters 6 and 10

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3.1 Circuits Containing Energy Storage Elements

AC circuit.

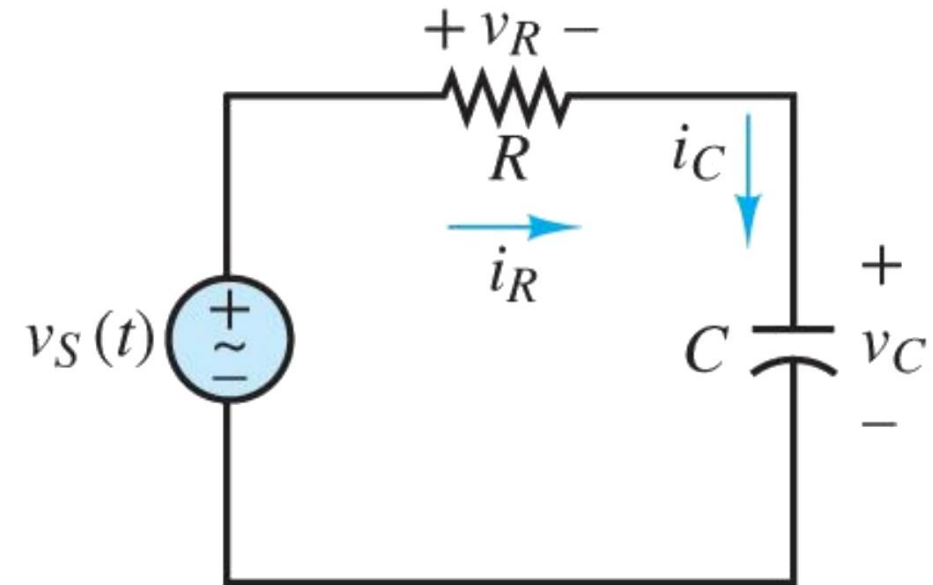
- Circuit with a sinusoidal source.

AC circuit with capacitors or inductors challenging.

- Require time to charge (store energy)
- i-v relationships are time dependent.

- A circuit containing energy storage elements is described by a differential equation.

$$\frac{di}{dt} + \frac{1}{RC}i = \frac{dv_S}{dt} \quad i_R = i_C = i$$



3.1 Circuits Containing Energy Storage Elements

AC circuit.

- Apply KVL around the loop to obtain:

$$v_S - v_R - v_C = 0$$

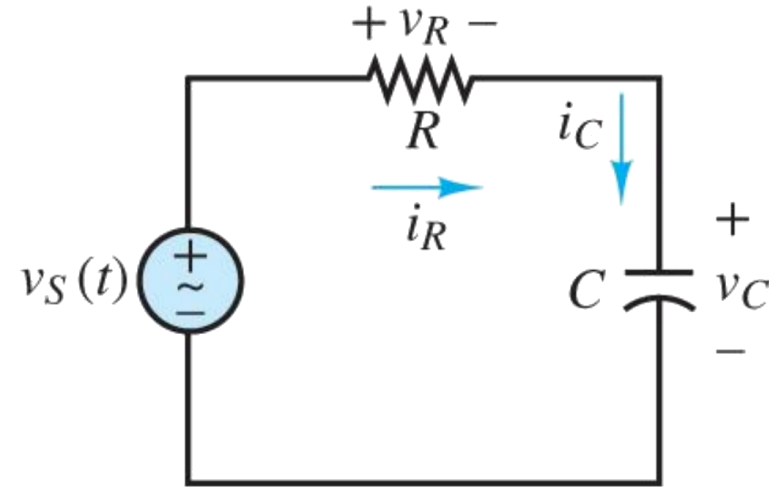
$$v_R = i_R R$$

$$i_C = C \frac{dv_C}{dt}$$

$$i_R = i_C = i$$

$$\frac{dv_C}{dt} + \frac{1}{RC} v_C = \frac{1}{RC} v_S$$

Time constant



(1st-order ODE)

Solution has two parts:

- (1) A transient solution (homogeneous)
- (2) A steady-state solution (particular)

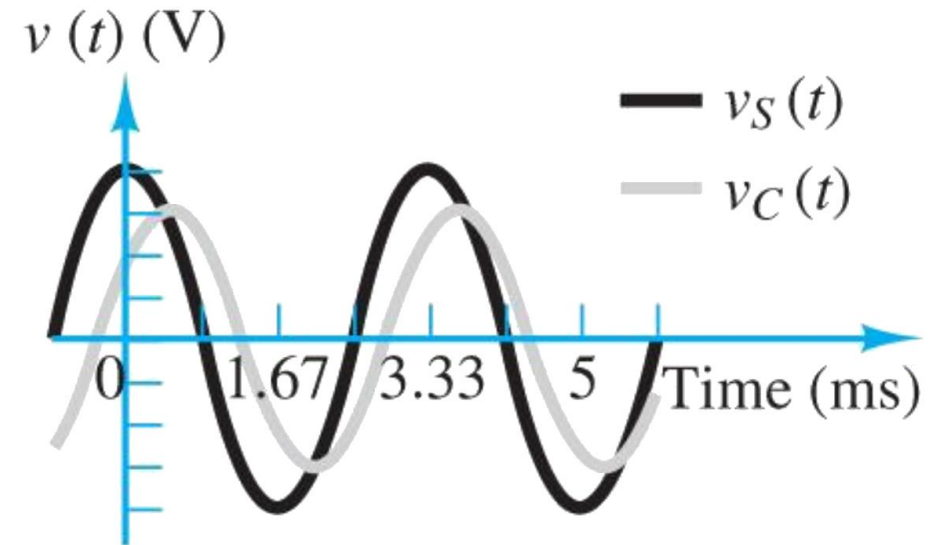
Alternative to Time Derivatives

Time delays are present in an AC circuit

- The delays are expressed as angles known as phase shifts

Solve for the steady state and transient solutions separately.

- Steady state solution explored in Chapter 3.
- Transient solution will be covered in Chapter 4.



3.2 Capacitors and Inductors

Capacitor.

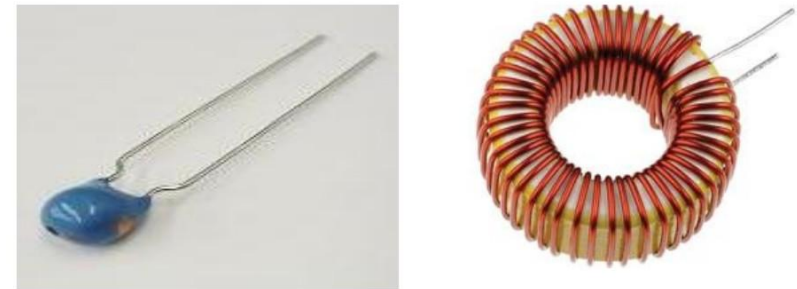
- Stores energy in electric field between two conducting plates.

Inductor.

- Stores energy in magnetic field of a conducting coil.

Leakage.

- Gradual reduction in stored energy over time.

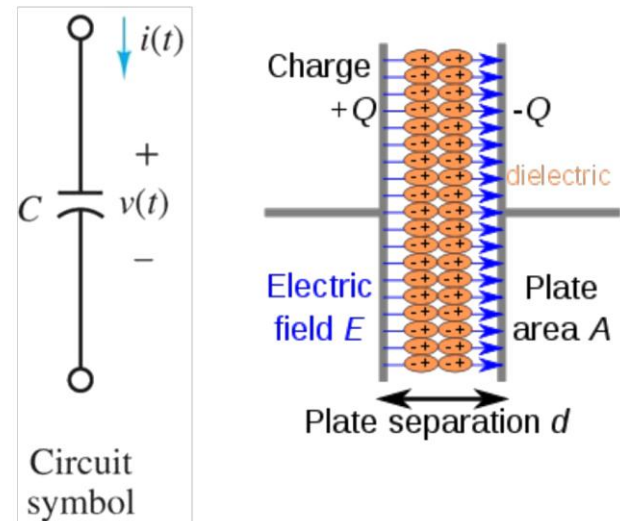
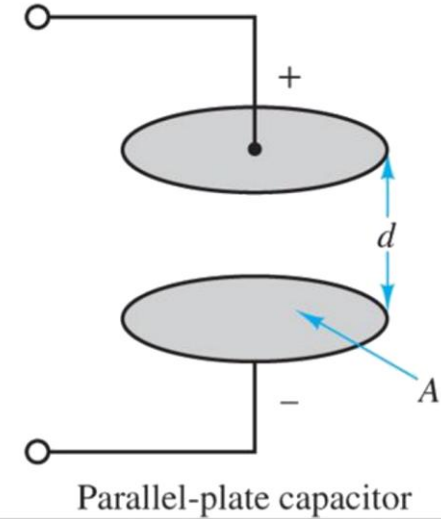
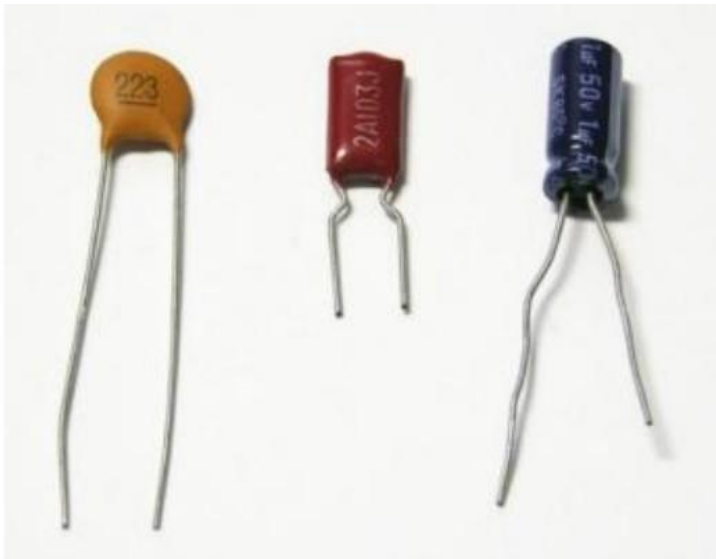


Properties of Capacitors and Inductors

	Capacitors	Inductors
Differential i - v	$i = C \frac{dv}{dt}$	$v = L \frac{di}{dt}$
Integral i - v	$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$	$i_L(t) = \frac{1}{L} \int_{-\infty}^t v_L(\tau) d\tau$
DC equivalent	Open-circuit	Short-circuit
Two in series	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Two in parallel	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
Stored energy	$W_C = \frac{1}{2} C v_C^2$	$W_L = \frac{1}{2} L i_L^2$

The Ideal Capacitor

- In a DC circuit, a capacitor is equivalent to an open circuit.
- Physical capacitors usually made with a dielectric material (air, mica, or Teflon) between the plates.



The Ideal Capacitor

- At all times the charge separation q_c is proportional to the applied voltage v_c

$$q_c = CV_c$$

- C is defined as a measure of the ability of the device to accumulate charge
- Unit: coulomb per volt, or farad (F)

$$i_c(t) = \frac{dq_c(t)}{dt}$$

$$i_c(t) = C \frac{dv_c(t)}{dt}$$

The Ideal Capacitor

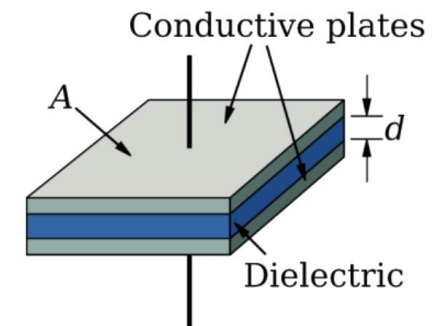
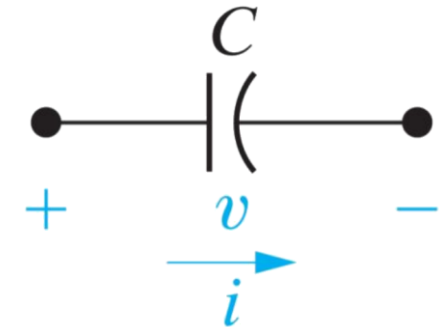
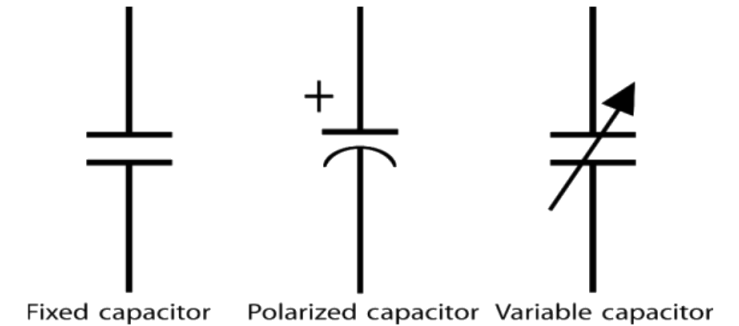
- Capacitance of an ideal parallel plate capacitor given by:

$$C = \frac{\kappa \epsilon_0 A}{d}$$

– κ : dielectric constant
 – ϵ_0 : permittivity constant (8.85×10^{-12} F/m)

- Charge separation is proportional to applied voltage.
- i-v relationship for capacitor given by:

$$i_C(t) = C \frac{dv_C(t)}{dt}$$



The Ideal Capacitor

- If the voltage drop across a capacitor is constant, the current is 0, so the capacitor can be replaced by an OPEN CIRCUIT.

Look at the capacitor equation again:

$$i_C(t) = C \frac{dv_C(t)}{dt}$$

- Suppose there is a discontinuity in $v(t)$, that is, at some value of t , the voltage jumps instantaneously. At this value of t , the derivative of the voltage is infinite. Therefore, the current is infinite! NOT POSSIBLE.
- **Thus, the voltage drop across a capacitor is continuous for all time.**

The Ideal Capacitor

- The equation for voltage in terms of current:

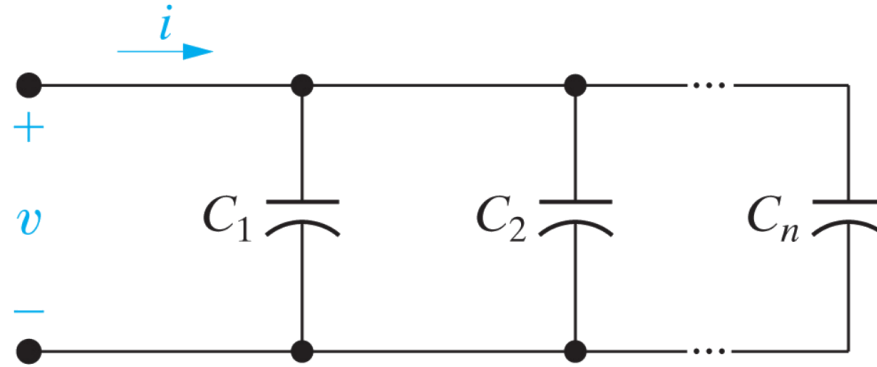
$$i(t) = C \frac{dv(t)}{dt} \quad \Rightarrow \quad i(t)dt = Cdv(t)$$

$$\Rightarrow \int_{t_o}^t i(\tau) d\tau = C \int_{v(t_o)}^{v(t)} dx$$

$$\Rightarrow v(t) = \frac{1}{C} \int_{t_o}^t i(\tau) d\tau + v(t_o)$$

Equivalent Capacitance

- Capacitors in parallel:

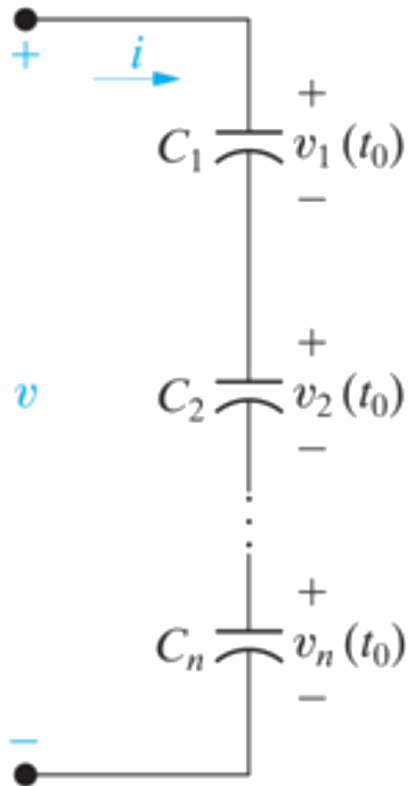


$$\begin{aligned}\text{KCL : } i &= i_1 + i_2 + i_3 = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} \\ &= (C_1 + C_2 + C_3) \frac{dv}{dt} = C_{eq} \frac{dv}{dt}\end{aligned}$$

Capacitors in parallel ADD.

Equivalent Capacitance

- Capacitors in series:



Capacitors combine in series like resistors or inductors combine in parallel.

$$\begin{aligned}
 \text{KVL: } v &= v_1 + v_2 + v_3 = \frac{1}{C_1} \int_{t_0}^t i(\tau) d\tau + v_1(t_0) \\
 &+ \frac{1}{C_2} \int_{t_0}^t i(\tau) d\tau + v_2(t_0) \\
 &+ \frac{1}{C_3} \int_{t_0}^t i(\tau) d\tau + v_3(t_0) \\
 &= \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) \int_{t_0}^t i(\tau) d\tau \\
 &\quad + [v_1(t_0) + v_2(t_0) + v_3(t_0)] \\
 &= \frac{1}{C_{eq}} \int_{t_0}^t i(\tau) d\tau + v_{eq}(t_0)
 \end{aligned}$$

Equivalent Capacitance and Energy Storage for Capacitors

Power and energy

$$p(t) = v(t)i(t) = Cv(t)\frac{dv(t)}{dt} \quad (\text{passive sign convention !})$$

$$p(t) = \frac{dw(t)}{dt} = Cv(t)\frac{dv(t)}{dt}$$

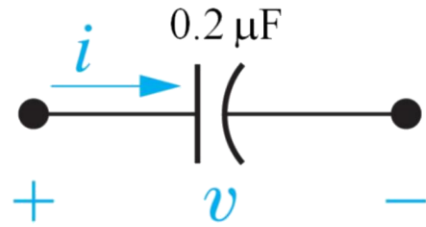
$$\Rightarrow dw(\tau) = Cv(\tau)dv(\tau)$$

$$\Rightarrow \int_0^{w(t)} dx = C \int_0^{v(t)} y(\tau) d\tau$$

$$\Rightarrow w(t) = \frac{1}{2} Cv(t)^2$$

Example 6.5: Capacitor Voltage, Power and Energy (1)

- Example 6.5: find the voltage, power, and energy



$$i(t) = \begin{cases} 0, & t \leq 0; \\ 5000t \text{ A}, & 0 \leq t \leq 20 \mu\text{s}; \\ 0.2 - 5000t \text{ A}, & 20 \leq t \leq 40 \mu\text{s}; \\ 0, & t \geq 40 \mu\text{s}. \end{cases}$$

For $t < 0$:

$$v(t) = 0 \text{ V}; \quad p(t) = 0 \text{ W}; \quad w(t) = 0 \text{ J}$$

For $0 \leq t \leq 20 \mu\text{s}$:

$$v(t) = \frac{1}{0.2 \mu} \int_0^t 5000 x \, dx + v(0) = \frac{1}{0.2 \mu} \frac{5000 x^2}{2} \bigg|_0^t = 12.5 \times 10^9 t^2 \text{ V}$$

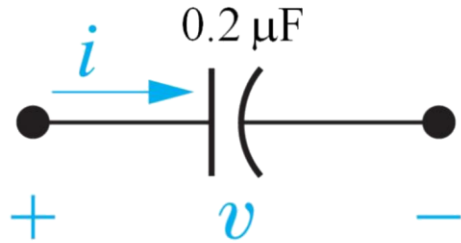
$$p(t) = v(t)i(t) = (12.5 \times 10^9 t^2)(5000 t) = 62.5 \times 10^{12} t^3 \text{ W}$$

$$w(t) = \frac{1}{2} (0.2 \mu)(12.5 \times 10^9 t^2)^2 = 15.625 \times 10^{12} t^4 \text{ J}$$

$$\text{At } t = 20 \mu\text{s}, \quad v(20 \mu\text{s}) = 12.5 \times 10^9 (20 \mu)^2 = 5 \text{ V}$$

Example 6.5: Capacitor Voltage, Power and Energy (2)

- Example 6.5, continued



$$i(t) = \begin{cases} 0, & t \leq 0; \\ 5000t \text{ A}, & 0 \leq t \leq 20 \mu\text{s}; \\ 0.2 - 5000t \text{ A}, & 20 \leq t \leq 40 \mu\text{s}; \\ 0, & t \geq 40 \mu\text{s}. \end{cases}$$

For $20 \mu\text{s} \leq t \leq 40 \mu\text{s}$:

$$v(t) = \frac{1}{0.2 \mu} \int_{20 \mu}^t (0.2 - 5000x) dx + v(20 \mu)$$

$$= \frac{1}{0.2 \mu} \left[0.2x - \frac{5000x^2}{2} \right]_{20 \mu}^t + 5$$

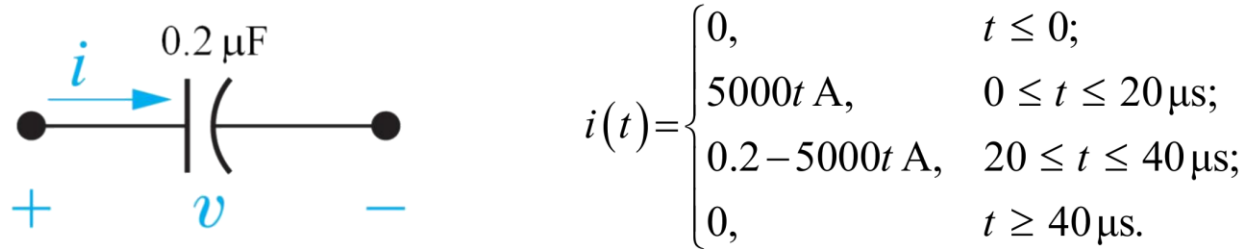
$$= (10^6 t - 12.5 \times 10^9 t^2 - 10) \text{ V}$$

$$p(t) = v(t)i(t); \quad w(t) = \frac{1}{2} C v(t)^2$$

$$\text{At } t = 40 \mu\text{s}, \quad v(40 \mu\text{s}) = [10^6 (40 \mu) - 12.5 \times 10^9 (40 \mu)^2 - 10] = 10 \text{ V}$$

Example 6.5: Capacitor Voltage, Power and Energy (3)

- Example 6.5, continued



For $t \geq 40 \mu\text{s}$:

$$v(t) = \frac{1}{0.2 \mu} \int_{40 \mu}^t 0 \, dx + v(40 \mu) = 10 \text{ V}$$

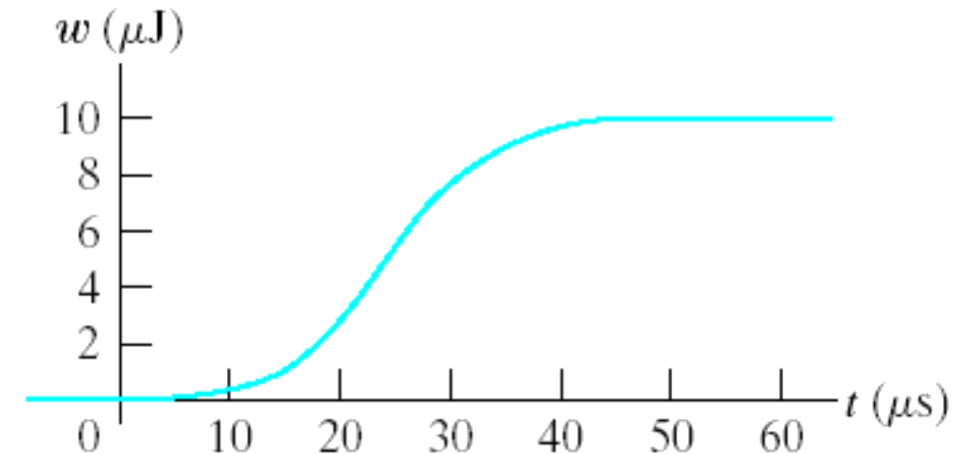
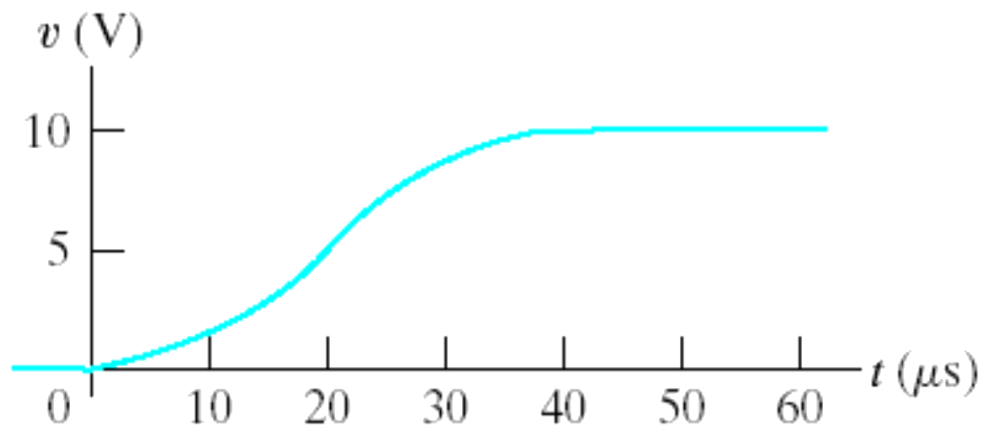
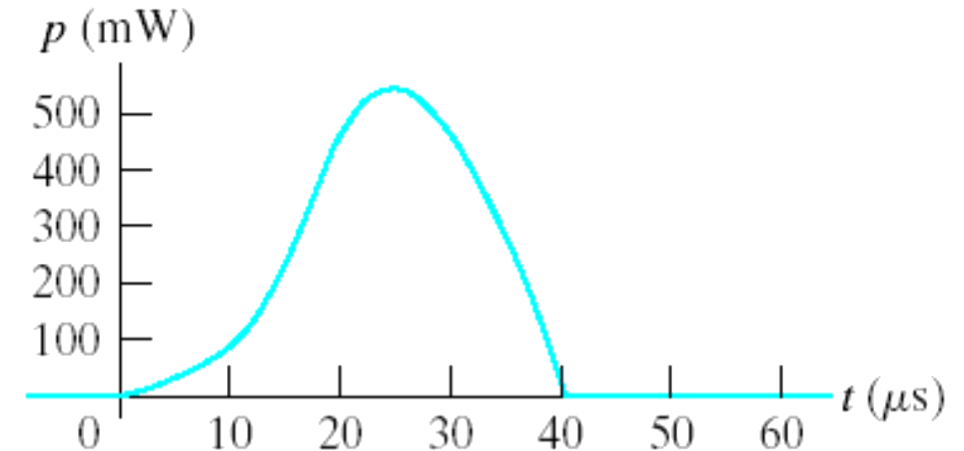
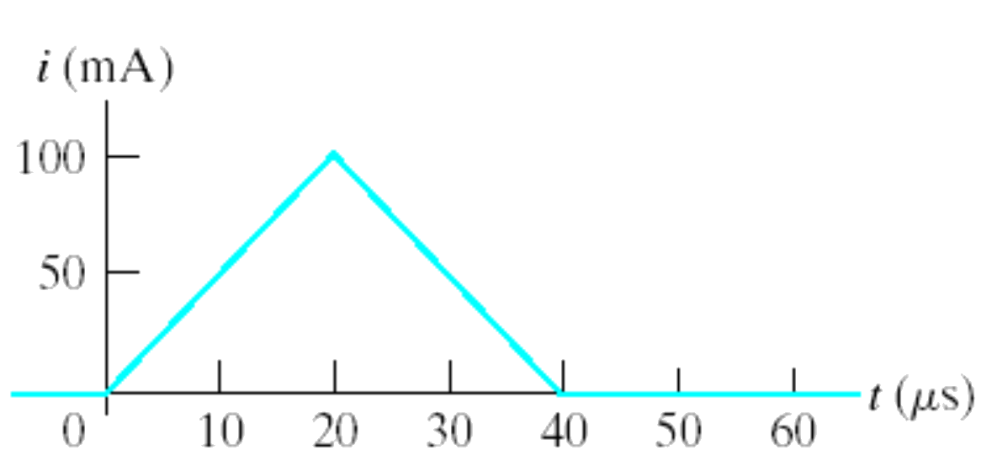
$$p(t) = v(t)i(t) = 0 \text{ W}$$

$$w(t) = \frac{1}{2}(0.2 \mu)(10)^2 = 10 \mu\text{J!}$$

During the interval between 0 and 40 μs , the power is positive (absorbed), energy is stored and “trapped” by the capacitor, so even when the current goes to 0, the voltage stays at 10 V and the energy is non-zero.

Example 6.5: Capacitor Voltage, Power and Energy (4)

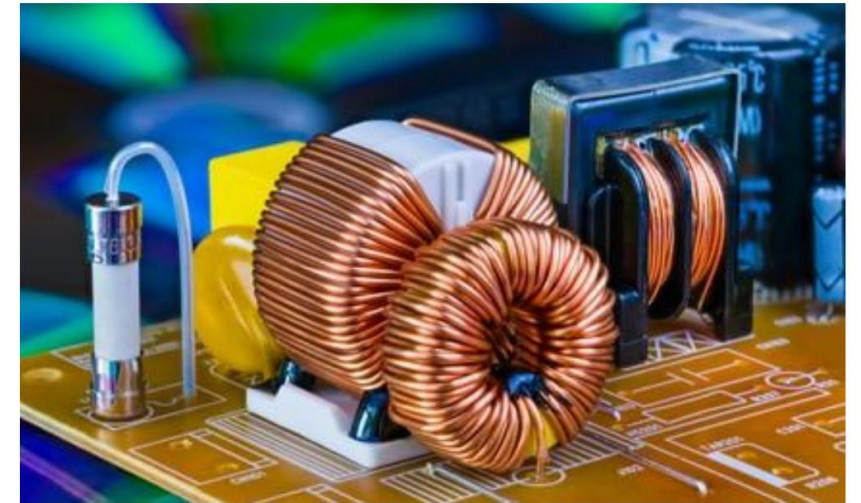
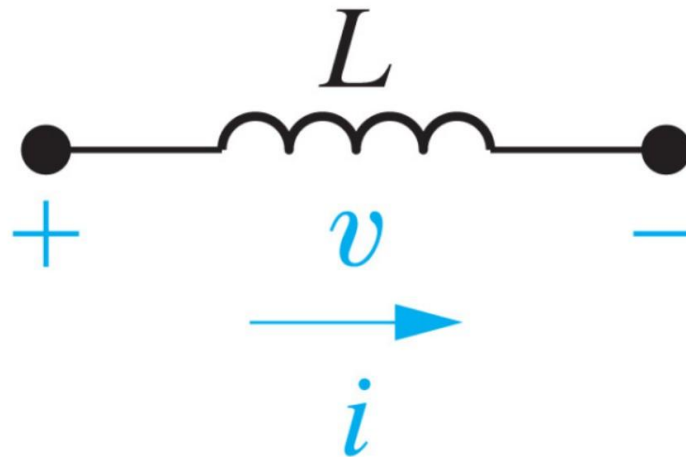
- Example 6.5, continued



The Ideal Inductor

Store energy in a magnetic field within and around a conducting coil

- Typically created by winding wire around a core.
- Inductor (inductance) present whenever a conducting wire is turned to form a loop.



The Ideal Inductor

- The inductance L is defined by:

$$L \equiv \frac{N\Phi}{i_L}$$

- Φ : magnetic flux through the core
- i_L : current through the inductor
- N : number of turns of the coil

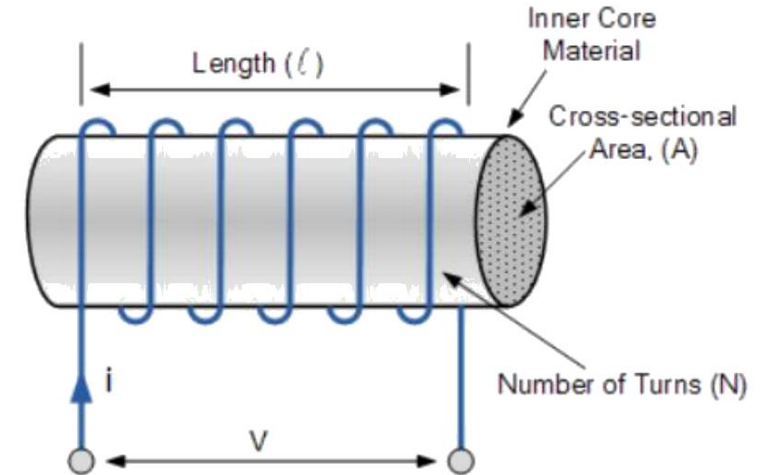
- The inductance of an ideal solenoid is:

$$L = \frac{\mu N^2 A}{l}$$

- μ : permeability of the core
- A : cross-section area of the coil
- l : axial length of the coil

- Unit: henrys (H), $1 \text{ H} = 1 \text{ V} \cdot \text{s/A}$
- i-v relationship from Faraday's law of induction

$$v_L(t) = L \frac{di_L(t)}{dt}$$



The Ideal Inductor

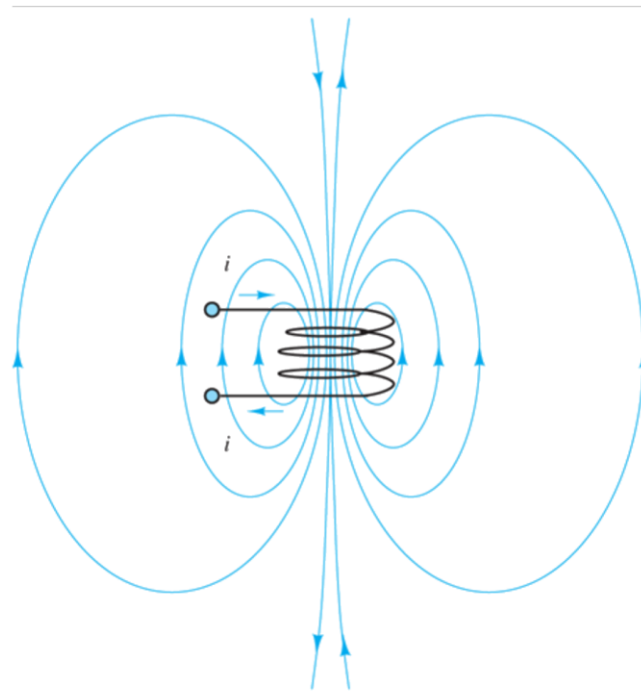
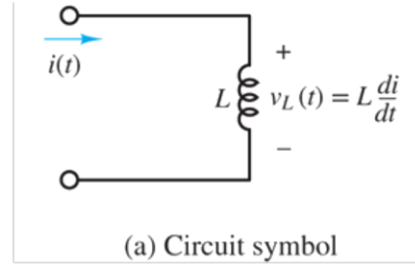
- If the current in the inductor is constant, the voltage is 0, so the inductor can be replaced by a SHORT CIRCUIT.

Look at the inductor equation again:

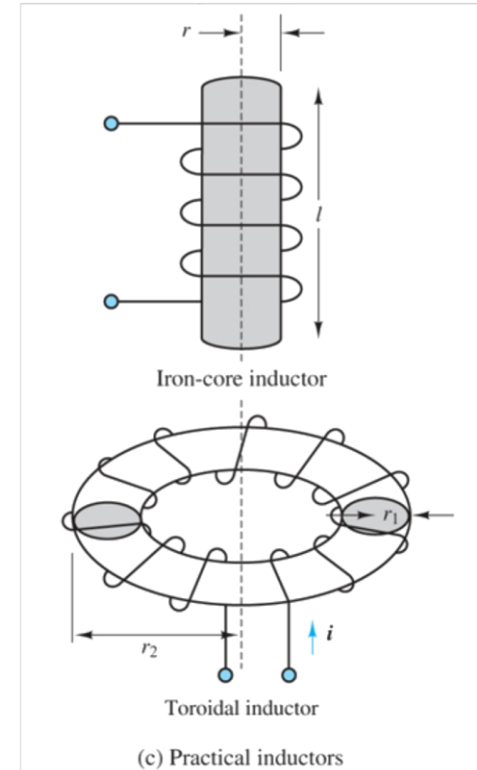
$$v_L(t) = L \frac{di_L(t)}{dt}$$

- Suppose there is a discontinuity in $i(t)$, that is, at some value of t , the current jumps instantaneously. At this value of t , the derivative of the current is infinite. Therefore the voltage is infinite! NOT POSSIBLE.
- **Thus, the current through an inductor is continuous for all time.**

Inductance and Practical Inductors



(b) Magnetic flux lines in the vicinity of a current-carrying coil



The Ideal Inductor

- The equation for current in terms of voltage:

$$v(t) = L \frac{di(t)}{dt} \quad \Rightarrow \quad v(t)dt = Ldi(t)$$

$$\Rightarrow \int_{t_o}^t v(\tau) d\tau = L \int_{i(t_o)}^{i(t)} dx$$

$$\Rightarrow i(t) = \frac{1}{L} \int_{t_o}^t v(\tau) d\tau + i(t_o)$$

The Ideal Inductor

- Power and energy:

$$p(t) = v(t)i(t) = Li(t)\frac{di(t)}{dt} \quad (\text{passive sign convention !})$$

$$p(t) = \frac{dw(t)}{dt} = Li(t)\frac{di(t)}{dt}$$

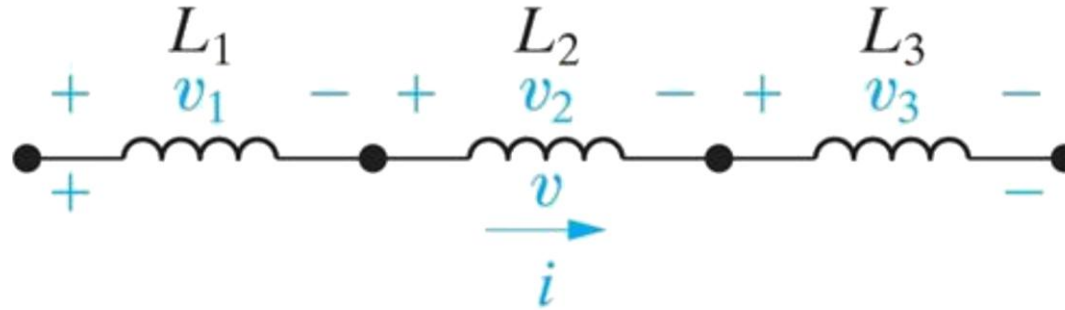
$$\Rightarrow dw(\tau) = Li(\tau)di(\tau)$$

$$\Rightarrow \int_0^{w(t)} dx = L \int_0^{i(t)} y(\tau) d\tau$$

$$\Rightarrow w(t) = \frac{1}{2} Li(t)^2$$

Equivalent Inductance

- Inductors connected in series:



$$v_1 = L_1 \frac{di}{dt}, \quad v_2 = L_2 \frac{di}{dt}, \quad \text{and} \quad v_3 = L_3 \frac{di}{dt}.$$

KVL:

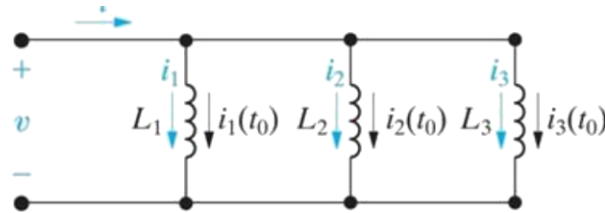
$$v = v_1 + v_2 + v_3 = (L_1 + L_2 + L_3) \frac{di}{dt},$$

Inductors in series ADD.

Equivalent Inductance

- Inductors connected in parallel:

Inductors combine in parallel like resistors combine in parallel.



$$i_1 = \frac{1}{L_1} \int_{t_0}^t v d\tau + i_1(t_0), \quad i_2 = \frac{1}{L_2} \int_{t_0}^t v d\tau + i_2(t_0), \quad i_3 = \frac{1}{L_3} \int_{t_0}^t v d\tau + i_3(t_0).$$

KCL: $i = i_1 + i_2 + i_3.$

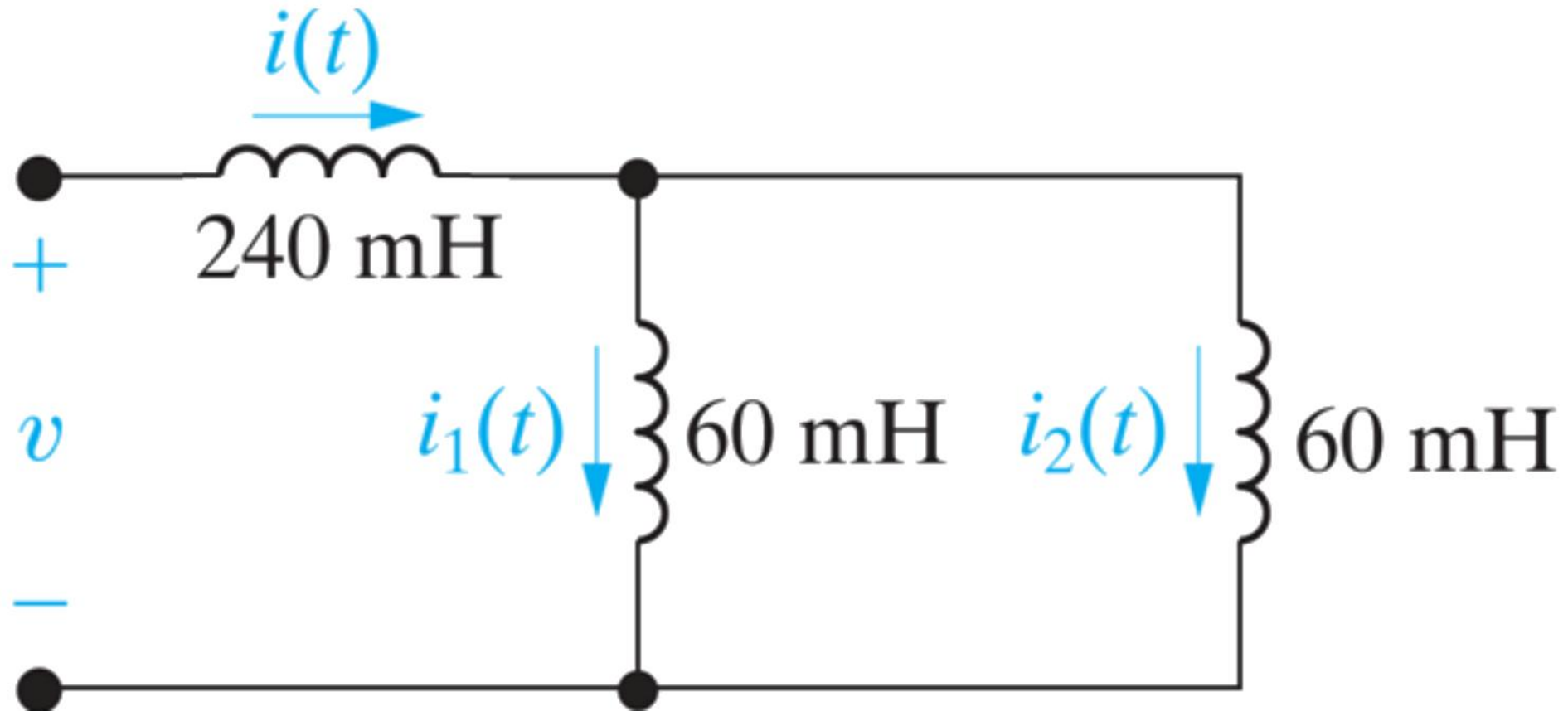
$$i = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right) \int_{t_0}^t v d\tau + i_1(t_0) + i_2(t_0) + i_3(t_0).$$

$$i = \frac{1}{L_{eq}} \int_{t_0}^t v d\tau + i(t_0).$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \quad i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

Example: Equivalent Inductance

- Find the equivalent inductance for the circuit below, assuming $i_1 = 6$ A and $i_2 = -3$ A.



3.3 Time-Dependent Sources

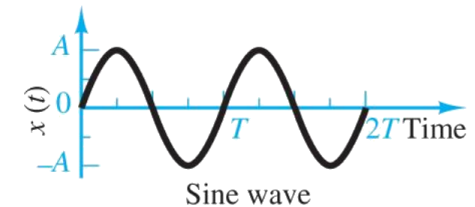
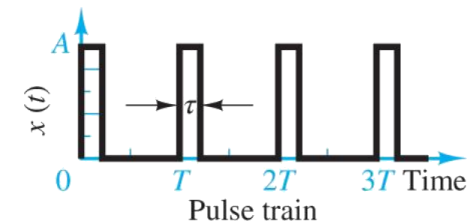
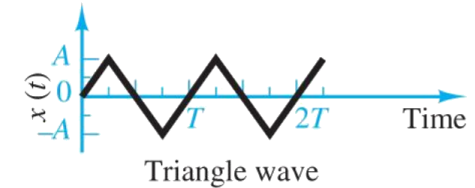
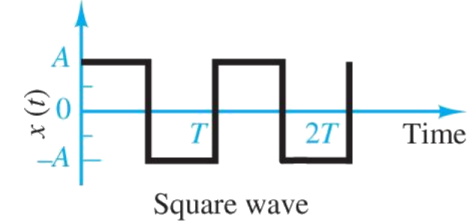
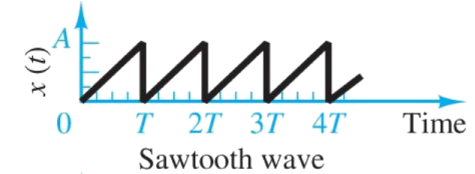
Electric power delivered in the form of periodic voltages and currents.

Sinusoids.

- Most important time-dependent waveform class.

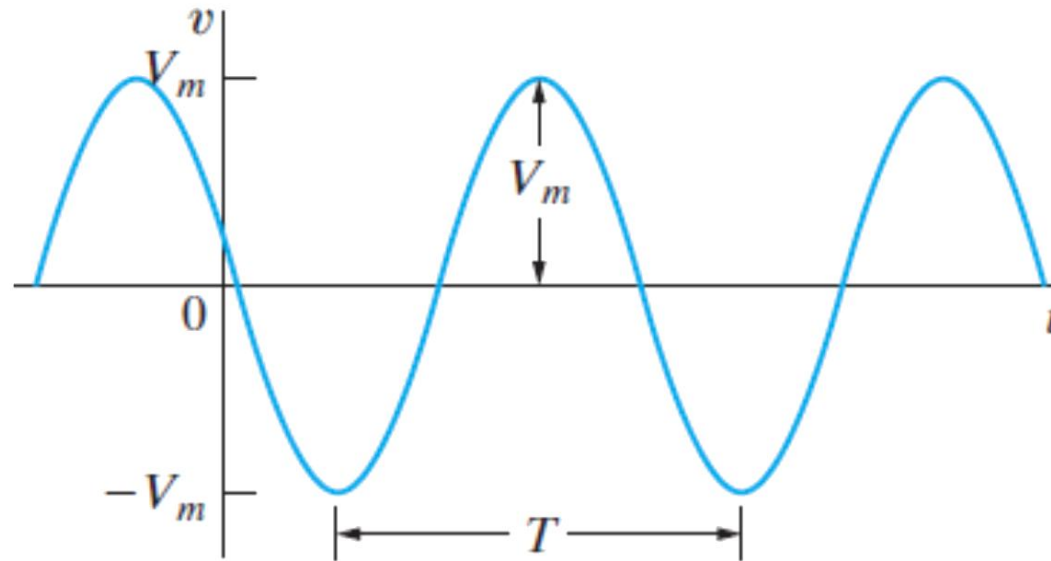
$$x(t) = A \cos(\omega t + \phi)$$

$$\omega = 2\pi f$$



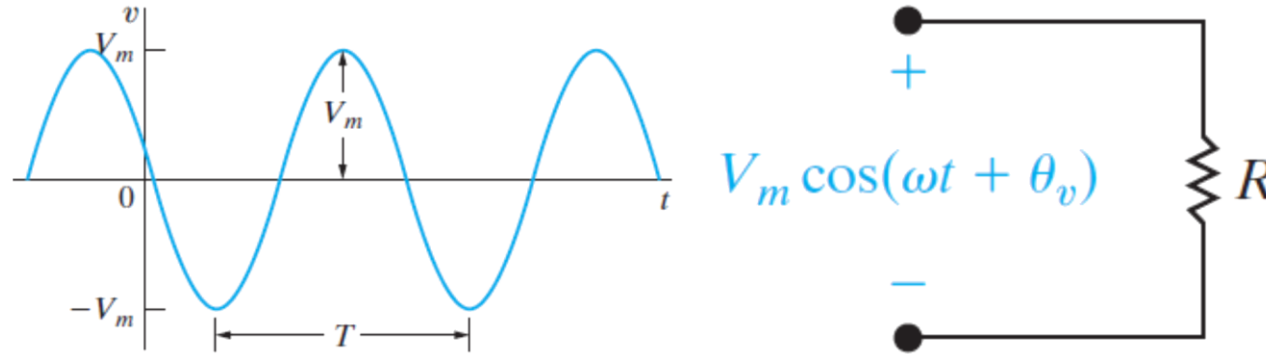
Effective or rms Values

- Effective or Root-mean-square (rms) value of a sinusoidal waveform.



$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt.}$$

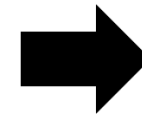
Effective or rms Values



- rms value of an ac source is the DC value that produces the same average power to be dissipated by a common resistor.

$$P = \frac{1}{T} \int_{t_0}^{t_0+T} \frac{V_m^2 \cos^2(\omega t + \phi_v)}{R} dt$$

$$= \frac{1}{R} \left[\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi_v) dt \right]$$



$$P = \frac{V_{\text{rms}}^2}{R}$$

$$P = I_{\text{rms}}^2 R$$